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# An integration of intelligent approaches and economic criteria for predictive analytics of occupational accidents ☆

Kamran Gholamizadeh [a], Esmaeil Zarei [b,c], Mohammad Yazdi [d,e], Matilde A. Rodrigues [f,g], Nasrin shirmohammadi-Khorram [h], Iraj Mohammadfam [i,*]

[a] Department of Occupational Health and Safety Engineering, Occupational Health and Safety Research Center, School of Public Health, Hamadan University of Medical Sciences, Hamadan, Iran
[b] Department of Safety Science, College of Aviation, Embry-Riddle Aeronautical University, Prescott, AZ, 86301, USA
[c] Robertson Safety Institute (RSI), Embry-Riddle Aeronautical University, Prescott, AZ, 86301, USA
[d] School of Engineering, Faculty of Science and Engineering, Macquarie University, Sydney, NSW, Australia
[e] School of Computing, Engineering & Physical Sciences, University of the West of Scotland (UWS), London, E14 2BE, UK
[f] Center for Translational Health and Medical Biotechnology Research (T-Bio), School of Health (ESS) of Polytechnic of Porto n° 400, 4200-072, Porto, Portugal
[g] Algoritmi Centre, School of Engineering, University of Minho, 4800-058 Guimarães, Portugal
[h] Department of Biostatistics, School of Public Health, Hamadan University of Medical Sciences, Hamadan, Iran
[i] Department of Ergonomics, Health in Emergency and Disaster Research Center, University of Social Welfare and Rehabilitation Science, Tehran, Iran

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ABSTRACT

Occupational accidents are a significant concern, resulting in human suffering, economic crises, and social issues. Despite ongoing efforts to comprehend their causes and predict their occurrences, the use of machine learning models in this domain remains limited. This study aims to address this gap by investigating intelligent approaches that incorporate economic criteria to predict occupational accidents. Four machine learning algorithms, Random Forest (RF), Support Vector Machine (SVM), Multivariate Adaptive Regression Spline (MARS), and M5 Tree Model (M5), were employed to predict occupational accidents, considering three economic criteria: basic income (BI), inflation index (II), and price index (PI). The study focuses on identifying the most suitable model for predicting the frequency of occupational accidents (FOA) and determining the economic criteria with the greatest influence. The results reveal that the RF model accurately predicts accidents across all income levels. Additionally, among the economic criteria, II had the most significant impact on accidents. The findings suggest that a reduction in FOA is unlikely in the coming years due to the increasing growth of II and PI, coupled with a slight annual increase in BI. Implementing appropriate countermeasures to enhance workers' economic welfare, particularly for low-income employees, is crucial for reducing occupational accidents. This research underscores the potential of machine learning models in predicting and preventing occupational accidents while highlighting the critical role of economic factors. It contributes valuable insights for scholars, practitioners, and policymakers to develop effective strategies and interventions to improve workplace safety and workers' economic well-being.

## 1. Introduction

Occupational accidents have emerged as a pressing global concern, garnering significant attention from the international community due to their profound impact on public health in the current century [1–3]. These accidents often result in fatalities, physical disabilities, loss of productivity, medical expenses, and substantial economic burdens [4–6]. Notably, in 2020, the European Union witnessed 2.7 million non-fatal work-related accidents and 3355 fatal accidents, indicating a ratio of approximately 815 non-fatal accidents for each fatal accident [7]. These statistics further reveal that there were 1.77 fatal accidents and 1444 non-fatal accidents per 100,000 individuals in the European Union during the same period [7]. Moreover, occupational accidents rank as the second leading cause of death in Iran [8]. In addition, Joghataei et al. (2023) [9] demonstrated that the disability-adjusted life years and deaths related to occupational injuries in Iran significantly are 169 and 523 in 2011 and 86 and 235 in 2018, respectively.

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* Corresponding author.
*E-mail addresses:* kamrangholamizade1373@gmail.com (K. Gholamizadeh), ezarei@erau.edu (E. Zarei), mohammad.yazdi@mq.edu.au, mohammad.yazdi@uws.ac.uk (M. Yazdi), mar@ess.ipp.pt (M.A. Rodrigues), n.shirmohamadi.kh@gmail.com (N. shirmohammadi-Khorram), Ir.mohammadfam@uswr.ac.ir (I. Mohammadfam).

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Moreover, it is essential to acknowledge that while various factors contribute to the occurrence of occupational accidents, economic crises in developing countries can significantly amplify the risk and frequency of such incidents (FOA) [10]. This holds particularly true for countries like Iran, where economic challenges have escalated in the past decade, primarily driven by heavy economic sanctions. These adverse economic conditions, characterized by soaring inflation rates, escalating price indices, and inadequate basic incomes for workers across diverse industries, have created a perilous environment [11]. The combination of these factors exacerbates the vulnerability of workers and raises concerns about the prevalence and severity of occupational accidents within the country. Understanding the intricate relationship between economic crises and occupational accidents is crucial for devising effective interventions and policies that address both the economic well-being of workers and their safety in the workplace. Therefore, this study aims to investigate the impact of economic factors on the frequency of occupational accidents in Iran, highlighting the urgent need for comprehensive strategies to mitigate the risks associated with economic crises and promote a safer working environment for all individuals.

The central hypothesis underlying this study posits that economic indicators exert a significant influence on the FOA. Economic crises hold tremendous sway over the lives of Iranians, and numerous studies have demonstrated the crucial role played by the inflation rate in shaping the quality of life of Iranian workers [12,13]. However, it is crucial to recognize that the impact of economic crises extends beyond workers' quality of life and permeates their working conditions within the workplace [14,15]. Research indicates that during favorable economic conditions, low income can have a profound effect on job satisfaction, enhancing employee motivation, fostering compliance with workplace regulations, and consequently reducing the occurrence of occupational accidents [16,17]. Thus, this study aims to analyze data on occupational accidents, which serve as a robust indicator of safety conditions in industries, with a specific focus on the fluctuations of the economic crisis. By investigating the relationship between economic indicators and FOA, this study seeks to provide valuable insights that can inform the development of targeted interventions and policies to enhance workplace safety in the face of economic challenges.

Despite the prevalence of occupational accidents across various industries, there remains a notable gap in comprehensive research focused on predicting the frequency of occupational accidents by incorporating changes in economic indicators. Conducting such a study holds immense potential in forecasting the impact of economic indicators on accident rates, thereby serving as a valuable tool for proactively controlling and managing work-related accidents in the future. This proactive approach can be utilized as a fundamental criterion for governing industrial operations within countries [18], aiding in the formulation of appropriate intervention strategies. Furthermore, this research aligns with the United Nations Sustainable Development Goals, specifically Goal 8 — decent work and economic growth. By integrating economic indicators into the prediction of FOA, this study contributes to the overarching aim of fostering safe and conducive working environments while driving sustainable economic development.

By bridging the gap between occupational accident prediction and economic indicators, this research offers significant practical implications. The findings can empower policymakers, industry stakeholders, and regulatory bodies to make informed decisions and implement targeted measures to reduce the frequency of accidents, enhance workplace safety, and improve workers' overall well-being. Moreover, this study serves as a stepping stone towards a more proactive approach to occupational safety management, facilitating early intervention and prevention strategies that prioritize the reduction of accidents based on economic indicators. Such proactive measures are crucial in promoting sustainable industrial development while ensuring the safety and welfare of workers, thereby contributing to the broader global agenda of achieving sustainable economic growth and decent work conditions.

Studying the impact of economic conditions on occupational accidents necessitates the utilization of advanced and precise analytical tools. The application of advanced statistical methods enables the evaluation of economic indicators' influence on occupational accidents and facilitates the prediction of future accidents based on accident reports and the effects of these parameters [19]. The primary reason for employing such methods lies in their ability to explore the intricate and interactive effects of various factors during accident investigations [20]. While numerous statistical methods exist for occupational accident prediction, their operational accuracy is often limited due to a lack of adequate flexibility. Traditional statistical approaches, in particular, tend to diminish the study's predictive accuracy when it comes to FOA, primarily due to the high level of uncertainty in bulk data analysis [21]. Consequently, there arises a pressing need to leverage advanced statistical methods in this context.

In contrast, the utilization of Machine Learning (ML) techniques offers a distinct advantage in predicting accidents [22]. ML approaches exhibit significant strengths over traditional statistical methods, owing to their inherent capacity for adaptability, flexibility, and precise prediction [19]. ML algorithms have the capability to discern complex patterns, capture non-linear relationships, and handle vast volumes of data, thereby facilitating accurate and reliable predictions of FOA. By harnessing the power of ML, researchers can unlock valuable insights hidden within occupational accident data, enabling them to develop robust models capable of accurately forecasting future accident frequencies based on economic indicators. The adoption of advanced statistical methods and ML approaches represents a pivotal advancement in the field of occupational accident prediction, promising enhanced precision and a deeper understanding of the underlying dynamics between economic conditions and accident occurrence [19].

The increasing inflation rate and the subsequent decline in purchasing power, particularly among low-income workers, have ignited concerns regarding the occurrence of occupational accidents. In light of this, predicting the FOA assumes paramount importance as it provides a roadmap for government bodies and managers to implement targeted countermeasures. The primary contribution of this study lies in the comparative analysis of various machine learning algorithms for predicting the incidence of occupational accidents within a large statistical population while considering the influence of economic indicators.

The present study seeks to address several key research questions to shed light on the complex interplay between economic indicators and occupational accidents:

(a) What has been the trend of occupational accidents over the past decade across different income levels of workers?
(b) Which economic criteria exert a significant impact on the occurrence of occupational accidents?
(c) What is the projected trajectory of occupational accidents based on economic criteria for the next five years?
(d) Among the economic criteria, which one holds the most substantial influence on occupational accidents?
(e) Which ML model demonstrates the highest efficacy in predicting FOA compared to other available models?

By answering these crucial questions, this study aims to provide valuable insights that can inform policymakers, government agencies, and industry stakeholders about the dynamics of occupational accidents concerning economic indicators. The findings will aid in formulating evidence-based strategies and intervention plans to mitigate the risks associated with occupational accidents and promote a safer working environment for individuals across diverse income levels. Additionally, identifying the most suitable machine learning model for predicting work-related accidents will contribute to the development of accurate and reliable predictive tools that can be employed in real-world settings to enhance occupational safety and achieve sustainable economic growth.

The rest of this paper continues as follows. Section 2 offers a comprehensive literature review that explores the application of ML techniques in predicting occupational accidents and investigates the impact of economic crises on accident frequencies. Section 3 delves into the intricacies of criteria extraction and provides a detailed description of the prediction models employed in this study. Moving forward, Section 4 presents the key findings encompassing the extracted criteria, job classification based on income levels, FOA prediction, and an evaluation of the models' reliability. In Section 5, we engage in in-depth discussions regarding the methodologies employed and compare our findings with relevant studies in the field. Finally, in the concluding section, we summarize the main implications drawn from this research and propose avenues for future research. Through this comprehensive exploration, our study aims to contribute to the growing body of knowledge surrounding the utilization of machine learning methods in predicting occupational accidents and sheds light on the critical role played by economic indicators in shaping workplace safety.

## 2. Literature review and research gaps

### 2.1. Studies related to machine learning and occupational accidents

The existing body of literature on the application of machine learning in predicting occupational accidents predominantly revolves around the construction industry, as evidenced by the studies presented in Table 1. However, a notable challenge encountered in these studies is the utilization of small statistical samples, which can potentially limit the generalizability of the findings. While previous research has primarily focused on the application of machine learning and artificial intelligence methods in analyzing and predicting occupational accidents, there appears to be a lack of specialized studies that undertake a comparative analysis of these methods specifically within the context of occupational accidents.

Furthermore, it is worth noting that the majority of studies have concentrated on the severity of accident outcomes, rather than the prediction of the actual number of accidents. This discrepancy suggests a gap in the existing research landscape. By recognizing these gaps, this current study aims to bridge the knowledge divide by examining and comparing various machine learning algorithms for predicting the occurrence of occupational accidents. Moreover, this research seeks to address the prediction of accident frequency, offering valuable insights into the future trends of occupational accidents based on economic criteria.

The literature review was expanded and updated to elucidate how the research contributes to the field of analytics, with a specific focus on the prediction and analysis of occupational accidents at a national level. While previous research has primarily concentrated on the application of machine learning techniques in accident prediction, the limitations of these studies have been acknowledged, such as their reliance on small statistical samples and the limited generalizability of findings. A gap in the literature has been identified in terms of conducting a comprehensive comparative analysis of various machine learning algorithms for the prediction of national-level occupational accidents. This research aims to bridge this gap by providing a detailed examination of these machine-learning methods. By doing so, not only are accidents predicted, but also the analytical framework is broadened to encompass the national scale, enabling more informed decisions regarding accident frequency, resource allocation, and proactive safety measures. Consequently, the study makes a significant contribution to the field of analytics by extending its application to the crucial domain of national-level occupational accident prediction, with far-reaching implications for both societal and economic considerations.

The present study takes on a predictive perspective, focusing on the frequency of occupational accidents and their intricate interactions with various economic criteria, enabling proactive risk management and resource allocation. Moreover, this study introduces a distinctive comparative element by comprehensively evaluating different machine learning algorithms, aiding the community in selecting the most effective approach for analyzing national-level occupational accidents. This combined effort enhances decision-making within the field, aligning with the broader trajectory of analytics where predictive and comparative analyses converge to empower stakeholders with valuable insights for complex decision environments.

### 2.2. Studies related to the impact of the economic criteria on accidents

The existing literature review reveals that studies exploring the impact of economic crises on accidents have primarily focused on the transportation sector. For instance, Liao et al. (2022) [23] compared occupational accidents between non-standard workers (workers who are employed with low daily wages) and standard workers in the construction industry and showed that the accidents rate and resulting injuries in non-standard workers are higher than standard workers and they attributed this difference to poor job security and protection in non-standard workers. In addition, Koc and Gurgun (2022) [24] introduced income, along with factors such as age and experience, as the most influential factors related to the severity of occupational accidents.

Furthermore, Amoadu et al. (2023) [25] highlighted that professional drivers in low- and middle-income countries (LMICs) face precarious working conditions, including high job demands, solo driving, irregular shifts, long hours, and limited job control. Factors such as job insecurity, effort-reward imbalance, and performance-based pay contribute to risky driving situations. Their research underscores the significant impact of these psychosocial work factors on driving performance in LMICs, emphasizing the connection between economic circumstances, workplace conditions, and accident rates on the road. The economic situation can be seen even in transport drivers. A study by Yang et al. (2023) [26] in Guangzhou, China showed that from 2017 to 2020, vulnerable road users, such as motorized two-to-three-wheeler drivers and migrant workers, were casualties in more than 80% of the cases. This illuminates how economic factors manifest in the safety and well-being of drivers, accentuating the complex relationship between economic conditions and the vulnerability of individuals on the road.

In addition, there remains a significant research gap in terms of specialized studies examining the influence of economic crises on occupational accidents. This knowledge gap constitutes the third gap in past studies.

Building upon these insights, the present study has been designed to address the three identified gaps. Previous studies have predominantly concentrated on predicting the frequency of occupational accidents using common machine learning methods, without conducting a comparative analysis of their effectiveness. Thus, the present study aims to fill this gap by comparing the performance of different machine learning models in predicting the occurrence of occupational accidents.

Moreover, while valuable studies have been conducted in this field, there is a need for further research to explore additional gaps. Previous studies have primarily focused on specific industries, and more comprehensive and accurate findings could be achieved by including a more diverse population from various workplaces. Furthermore, when examining economic indicators and income, it is essential to consider complementary factors such as inflation rate and price index. The integration of economic indicators with occupational accidents has not yet been thoroughly investigated in specialized studies. Additionally, there is a lack of quantitative comparisons of machine learning models in analyzing occupational accidents and monitoring economic indicators, which are crucial influencing factors that have been overlooked in previous research.

These challenges related to occupational accidents and the economic crisis faced by workers form the basis of the present research. The study aims to address these existing challenges by predicting the frequency of occupational accidents in the coming years while also

**Table 1**
Studies conducted in the field of the application of machine learning in predicting occupational accidents.

| Study | Field of the study | Methods used | Comparison between methods |
|---|---|---|---|
| Sarkar et al. (2019) [19] | Predicting accidents and near-miss | Support Vector Machines (SVM), Artificial Neural Networks (ANN) | Yes |
| Kang et al. (2019) [27] | Predicting constructions accidents | Random Forest (RF) | No |
| Zhu et al. (2021) [28] | Predicting the severity of outcomes in construction accidents | Decision Tree (DT), SVM, Logistic Regression (LR), Naïve Bayes, K-nearest Neighbor (KN), Multi-layer preceptor (MLP), Auto-ML | Yes |
| Yedla et al. (2020) [29] | Predicting the severity of safety outcomes in mining operations | DT, RF, ANN | Yes |
| Zaranezhad et al. (2019) [30] | Predicting the maintenance-related accidents | ANN, Fuzzy System, Genetic Algorithm, Colony Optimization Algorithm | Yes |
| Lee et al. (2020) [31] | Predicting accidents in the constructions industry | SVM, DT | No |
| Kakhki et al. (2019) [32] | Predicting the severity of agribusiness industries | SVM, Boosted Tree, Naïve Bayes | Yes |
| Sarkar et al. (2020) [33] | Predicting the severity of outcomes | SVM, Naïve Bayes, RF, ANN, Classification and Regression Tree | Yes |
| Sarkar et al. (2016) [34] | Predicting occupational accidents | DT | No |
| Baker et al. (2020) [35] | Predicting the severity of outcomes in construction accidents | XGBoost, SVM | No |
| Lou et al. (2023) [1] | Predicting the severity of outcomes in occupational accidents | RF | No |
| Zermane et al. (2023) [36] | Predicting fatal fall from heights accidents | Extreme Gradient Boosting, DT, RF, MLP, KN, SVM, Logistic Regression (LR) | Yes |
| Gatera et al. (2023) [37] | Comparing between machine learning models performances in predicting traffic accidents | RF, SVM | Yes |
| Present study | Predicting occupational accidents in Iran based on economic criteria | SVM, RF, M5 model, Multivariate adaptive regression spline (MARS) | Yes |

comparing the superiority of various machine learning models. By doing so, the study endeavors to provide valuable insights that contribute to the prevention and management of occupational accidents, taking into account the impact of economic indicators. Ultimately, this research aims to improve the understanding of the complex relationship between economic factors and occupational accidents, enabling policymakers and stakeholders to implement effective interventions and strategies to enhance workplace safety and workers' economic well-being.

## 3. Materials and methods

The study consists of three main phases, as depicted in Fig. 1. In the first phase, data pertaining to occupational accidents and economic indicators are collected and meticulously classified. These data serve as the foundation for subsequent analyses. The second phase focuses on predicting the FOA utilizing ML models, leveraging the data gathered in the first phase. The ML models are employed to generate accurate predictions based on the identified patterns and trends in the dataset. Finally, in the third phase, the performance and effectiveness of the selected ML techniques are evaluated and assessed. This rigorous evaluation allows for the determination of the most suitable ML models for predicting FOA, ensuring the reliability and validity of the study's findings.

### 3.1. Collecting a data

In this study, a comprehensive dataset of accident data related to insured workers across all industries in Iran was collected from 2007 to 2017. The dataset consisted of a total of 191,879 accidents and was obtained from the comprehensive report of the Social Security Organization, a national organization that covers insured workers throughout the country. The collected information encompassed various details, including the worker's occupation, the date of the accident, the type and cause of the accident, and the number of days the worker was absent from work.

The accident data was further categorized based on the workers' income level. Each accident was assigned to one of three income categories: low-income (income below 2.5 million Tomans ≅ $100), medium-income (income between 2.5 million and 5 million Tomans ≅ $100 to $200), and high-income (income above 5 million Tomans ≅ >$200). This categorization allowed for a more nuanced analysis of the relationship between income and accident occurrence.

Alongside the accident data, economic indicators were also extracted for the period of 2007–2017. Three primary economic indicators were considered in this study: basic income (BI), price index (PI), and inflation index (II). These indicators were identified as influential factors affecting the lives of workers in previous studies [38] and were therefore included in the analysis.

The collected accident data and economic indicators served as the foundation for subsequent investigations and analyses conducted in this study. The relationship between these factors and their impact on occupational accidents was examined in detail, leading to valuable insights and findings.

### 3.2. Machine learning algorithms

#### 3.2.1. Random Forest (RF)

The Random Forest (RF) regression model is a powerful ensemble learning method that combines multiple decision trees to make predictions [39]. Each tree in the RF model is built through a process of recursive splitting of the input space into homogeneous subspaces, resulting in a collection of diverse trees [40]. One of the key strengths of the RF model is its ability to handle complex relationships and capture nonlinear effects and higher-order interactions among the predictors [41]. This is achieved through the combination of multiple

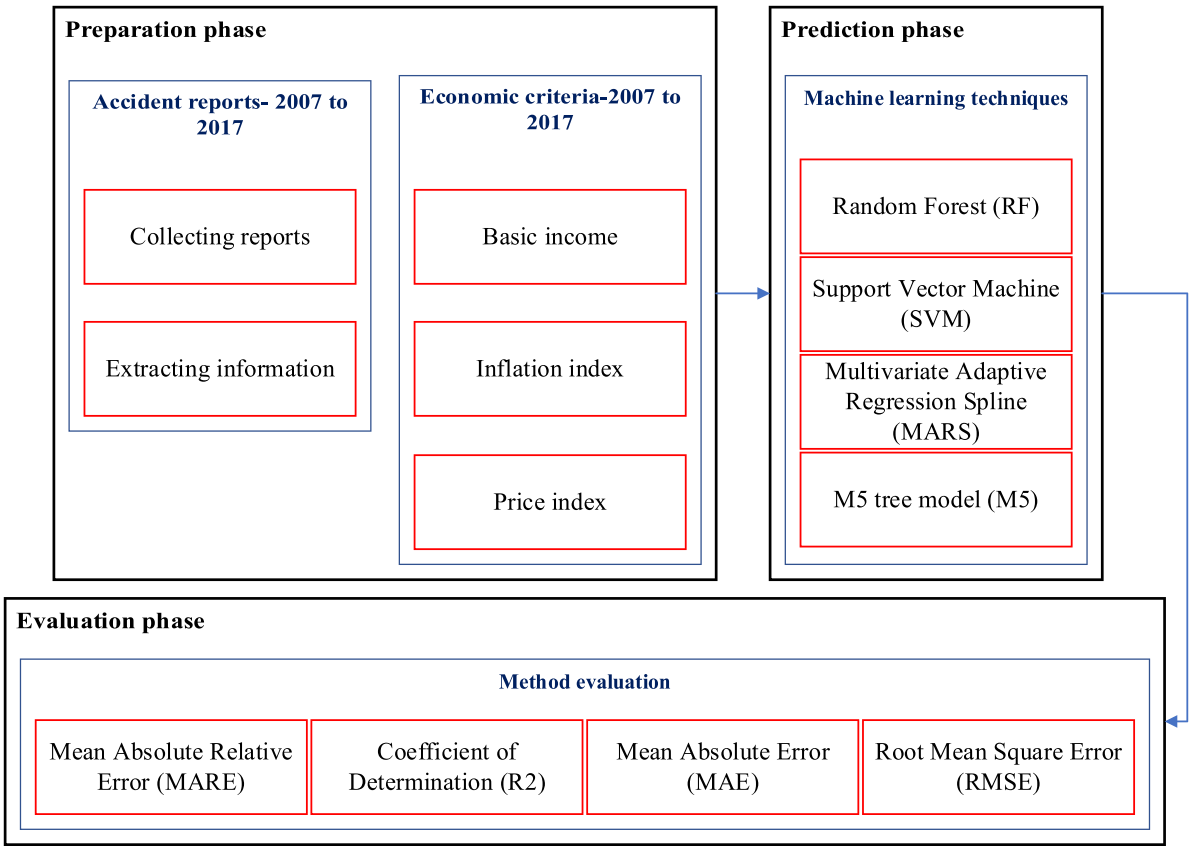


**Fig. 1.** The research phases.

trees, each trained on a different subset of the data and a random subset of predictors [42]. By aggregating the predictions from individual trees, the RF model produces a more robust and accurate overall prediction [43].

In the RF model, the importance of predictors can be assessed by examining their role in the creation of trees. Predictors that contribute significantly to the splitting decisions in the trees are considered to be the most important predictors [41]. This feature of the RF model allows for the identification of the key predictors that have the strongest influence on the frequency of occupational accidents (FOA) in our study.

To determine the fundamental predictors affecting the FOA, the influence of each predictor was evaluated. The rate of increment in forecast error was calculated by selectively withholding the data for a particular predictor while keeping the other predictors constant [43]. This analysis helps in understanding the relative importance of predictors and provides valuable insights into their contributions to the prediction of FOA. Relative importance (%IncMSE) is a statistical metric used in machine learning to assess the contribution of individual predictor variables to the mean squared error (MSE) of a predictive model. It quantifies the extent to which each variable influences the model's overall performance, with higher values indicating greater importance. This metric helps identify which features have the most significant impact on the model's accuracy, assisting in feature selection and understanding variable contributions [44]. For more detailed information on the Random Forest regression model, the study by James G et al. (2013) [45] is recommended as a comprehensive resource that delves into the intricacies and applications of RF in predictive modeling.

### 3.2.2. Support Vector Machine (SVM)

The Support Vector Machine (SVM) is a widely used method in statistical learning for classification and regression tasks [46,47]. In regression problems, the SVM technique aims to approximate the dataset in a high-dimensional space using a linear function [48]. This method employs a loss function that takes into account both the empirical error and a regularization norm, allowing for effective regularization and avoidance of overfitting [49,50]. By minimizing the expected error of the learning machine, SVM not only seeks to accurately fit the training data but also aims to enhance its generalization performance for prediction tasks [51]. This characteristic of SVM contributes to its ability to handle unseen data and make reliable predictions in real-world scenarios [52]. The function of SVM in regression tasks can be mathematically represented by Eq. (1) [53]:

$$y = k(x) = w \quad (x) + b \tag{1}$$

where:
x is the input vector
y shows the corresponding output value
w represents the weight vector
b is the bias
(x) indicates a nonlinear function in vector space.

SVM's effectiveness in handling regression problems lies in its ability to find an optimal hyperplane that maximizes the margin between different classes or approximates the data points with minimum error in the case of regression [51,54]. This optimization process allows this approach to achieve a robust and accurate regression model, capable of capturing complex relationships and achieving high predictive performance.

The SVM method has been widely employed in various domains due to its flexibility, generalization ability, and solid theoretical foundations [55–57]. Researchers have successfully utilized it in predicting various outcomes and it has demonstrated favorable performance in different applications. Therefore, it offers a valuable approach for our study in predicting the frequency of occupational accidents based on economic criteria, contributing to a better understanding of the relationship between these factors and enhancing workplace safety.

### 3.2.3. Multivariate adaptive regression spline (MARS)

Multivariate Adaptive Regression Spline (MARS) is a powerful nonlinear regression model widely used in statistical learning theory [58]. It offers a flexible approach for predicting continuous numeric outcomes and can be viewed as an enhancement of the classification and regression tree method [59]. One of the key advantages of this model is its nonparametric nature, which eliminates the need for making specific assumptions about the functional relationship between predictor and outcome variables [60,61]. This characteristic allows the MARS model to adaptively capture complex and nonlinear patterns present in the data, making it suitable for modeling real-world phenomena that may involve intricate interactions between variables.

The MARS model constructs a regression function using a series of basis functions known as splines, which are piecewise-defined polynomials. These splines are automatically generated and adapted to different regions of the predictor space based on the observed data. By locally adjusting the shape and complexity of the splines, this model effectively captures the underlying patterns in the data and provides a highly flexible and interpretable regression model. The general form of the regression function in MARS can be represented by Eq. (2) [62]:

$$y = {}_0 + \sum {}_i \max(0; x_i - v_i)^* + \sum_{i \neq j} {}_{ij} \max(0; x_i - v_i)^* \max(x_j - v_j)^* + \cdots \quad (2)$$

where:

0 shows the regression coefficients

$_i$ and $_{ij}$ represents some constant (known as knots).

The MARS model possesses several positive characteristics that make it a valuable tool in regression analysis. It can handle both numerical and categorical predictors, allowing for the inclusion of a wide range of variables in the model [63]. Moreover, it is capable of capturing interactions between predictors, detecting breakpoints in the relationships, and providing a clear representation of the nonlinear patterns in the data [64]. This interpretability makes this model not only a powerful prediction tool but also a useful tool for understanding the underlying mechanisms driving the outcomes of interest.

Given its flexibility and ability to handle complex relationships, the MARS model presents a promising approach for predicting the frequency of occupational accidents in our study. By incorporating economic indicators and their interactions, this model can effectively capture the nonlinear effects of these factors on the occurrence of accidents, thereby enhancing our understanding of the relationship between economic conditions and workplace safety.

### 3.2.4. M5 tree model (M5)

The M5 tree model (M5), introduced by Keshtegar et al. (2023) [65], is a machine-learning method that combines the advantages of binary decision support systems with linear regression in the leaf nodes to establish the relationship between predictor and outcome variables [66]. Unlike traditional regression tree learners, this model excels in simulating high-dimensional data, making it suitable for datasets with a large number of attributes [67].

The M5 model follows a two-phase process for constructing the regression tree. In the first phase, the data is partitioned into subsets using a splitting criterion, which leads to the creation of an initially overgrown tree. However, overgrown trees can be prone to overfitting, which may compromise the model's generalization ability. To address this, the second phase focuses on pruning the overgrown tree by replacing subtrees with linear regressions [68,69]. This pruning process helps simplify the model and enhances its interpretability without sacrificing its predictive accuracy. The overall form of this model can be represented by Eq. (3):

$$y = {}_0 + \sum_i {}_i \quad (3)$$

where:

$_i$ is linear regression coefficients [67].

The M5 tree model offers several positive characteristics that contribute to its effectiveness in regression analysis. It can handle both numerical and categorical predictors, allowing for a wide range of variables to be included in the model. Moreover, this model can capture both linear and nonlinear relationships between predictors and outcomes, providing a flexible framework for modeling complex data patterns. Additionally, the model's interpretability is enhanced by the linear regression models employed in the leaf nodes, which provide explicit equations that describe the relationship between predictors and outcomes [70].

In our study, the M5 tree model proves to be a valuable tool for predicting the frequency of occupational accidents. By leveraging its ability to handle high-dimensional data and incorporate both linear and nonlinear relationships, this model can effectively capture the complex interactions between economic indicators and accident occurrences. The interpretability of the model allows us to gain insights into the specific factors and their impact on occupational accidents, thereby facilitating the development of targeted interventions and preventive measures in the workplace.

### 3.2.5. Evaluating criteria

To evaluate and compare the forecast precision and efficiency of prediction models using the dataset in this study, several performance metrics were employed. These metrics include the Root Mean Square Error (RMSE), Mean Absolute Error (MAE), Coefficient of Determination (R2), and Mean Absolute Relative Error (MARE). These metrics provide standardized measures to assess the accuracy and effectiveness of the predictive models.

The RMSE is a widely used metric that quantifies the average magnitude of the differences between the observed and predicted values. A lower value indicates better forecast accuracy, as it signifies smaller deviations between the predicted and actual values [71]. Similarly, the MAE measures the average absolute difference between the predicted and observed values [72]. It provides a measure of the model's average prediction error and is also used to evaluate the model's accuracy. A lower value indicates a better forecast precision and a closer fit between the predicted and actual values [73].

In addition, the MARE is a relative error metric that measures the average absolute difference between the predicted and observed values, normalized by the mean of the observed values. It allows for the comparison of forecast accuracy across different scales and is particularly useful when dealing with datasets that have varying magnitudes. A lower value indicates a better forecast efficiency, as it indicates a smaller relative deviation between the predicted and observed values [74]. The coefficient of determination, R2, measures the proportion of the variance in the observed values that can be explained by the predicted values. It indicates the goodness-of-fit of the model and measures the agreement between the observed and forecast number of events. A higher value indicates a better agreement between the predicted and actual values, suggesting a stronger predictive performance of the model [75].

These performance metrics are computed using specific formulas, as represented by Eqs. (4), (5), (6), and (7), respectively. These metrics provide standardized measures for evaluating and comparing the forecast accuracy and efficiency of the models in predicting the frequency of occupational accidents in our study. In summary, by employing these performance metrics, we can objectively assess the predictive capabilities of the different machine learning models and identify the most accurate and efficient model for predicting the occurrence of occupational accidents based on economic indicators [43,76].

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^{N} (y_{observed} - y_{predicted})^2} \quad (4)$$

where:

$N$ is the total number of data points or observations.

$y_{observed}$ represents the actual values or observations.

$y_{predicted}$ represents the values predicted by the model.

The lower the RMSE, the better the model's predictive accuracy. It penalizes larger errors more significantly.

$$MAE = \frac{1}{N}\sum_{i=1}^{N} |y_{observed} - y_{predicted}| \quad (5)$$

where:

$N$ is the total number of data points or observations.

$y_{observed}$ represents the actual values or observations.

$y_{predicted}$ represents the values predicted by the model.

MAE provides a measure of the average magnitude of errors in the predictions. It is less sensitive to outliers compared to RMSE.

$$MARE = \frac{1}{N}\sum_{i=1}^{N} \left|\frac{y_{observed} - y_{predicted}}{y_{observed}}\right| \quad (6)$$

where:

$N$ is the total number of data points or observations.

$y_{observed}$ represents the actual values or observations.

$y_{predicted}$ represents the values predicted by the model.

MARE is useful when you want to understand the average relative error in percentage terms. It helps gauge the model's performance in a relative context.

$$R^2 = 1 - \frac{\sum_{i=1}^{N}(y_{predicted} - y_{mean})^2}{\sum_{i=1}^{N}(y_{observed} - y_{mean})^2} \quad (7)$$

where:

$N$ is the total number of data points or observations.

$y_{observed}$ represents the actual values or observations.\

$y_{predicted}$ represents the values predicted by the model.

$y_{mean}$ represents the mean of the observed values.

$R^2$ ranges from 0 to 1, where a higher value indicates a better fit of the model to the data. An $R^2$ value of 1 indicates a perfect fit.

Machine learning analysis was conducted using packages of R (version 3.4.0) software of "e1071" Version 1.7–3 [77], "Random Forest" version 4.6–14 [78], "earth" version 5.1.2, [79], and "RWeka" version 0.4–42 [80] which are publicly available.

## 4. Results

### 4.1. Economic indicators and accidents occurrence

Table 2 provides a comprehensive breakdown of the descriptive statistics for various parameters within three distinct income groups: Low income, Medium income, and High income, across three different datasets: the entire dataset spanning from 2007 to 2017, a training dataset encompassing the years 2007 to 2015, and a test dataset covering the years 2016 to 2017. The analysis revealed that the average II was 18.18 (standard deviation: 8.303), the average PI was 63.91 (standard deviation: 34.460), and the average BI was 602 217.769 Tomans (standard deviation: 387 887.057 Tomans). These findings provide a comprehensive overview of the economic indicators considered in this study.

Furthermore, the distribution of accidents among different income groups was examined. It was observed that more than half of all accidents (58%) occurred among low-income workers. This highlights the vulnerability of this particular group to occupational accidents and emphasizes the need for targeted interventions to improve workplace safety.

To gain further insights into the trend of FOA among different income groups, a graphical representation was generated (see Fig. 2). The trend analysis indicated a decrease in the FOA among both low- and medium-income workers during the study period from 2007 to 2017. However, it is important to note that the changing trend exhibited a nonlinear pattern, suggesting the presence of more complex dynamics that warrant further investigation.

These initial findings provide valuable insights into the relationship between economic indicators, income levels, and the occurrence of occupational accidents. The data highlights the concentration of accidents among low-income workers and the varying trends in accidents across different income groups. However, to fully understand the underlying factors contributing to these patterns, more in-depth studies are necessary. Further research should explore the complex relationships between economic indicators, income levels, and occupational accidents to uncover the underlying dynamics and inform the development of targeted strategies for accident prevention and improved workplace safety.

### 4.2. Predicting and evaluating results

#### 4.2.1. Machine learning algorithms' performance in predicting the FOA

The performance of models in terms of forecast precision and efficiency was evaluated using several metrics on the test sets. The results of this comparison are presented in Table 3. The findings indicate that the M5 model (RMSE: 3.19, MAE: 1.90, MARE: 0.04, and R2: 0.99) demonstrated the highest efficiency among the models, outperforming the others in terms of price index predictions. Additionally, the RF model (RMSE: 2.83, MAE: 2.15, MARE: 0.13, and R2: 0.93) exhibited superior performance in modeling and predicting the inflation index compared to the other models.

Moreover, the results highlight the RF model's efficacy in predicting the number of workers' accidents across different income levels. This model consistently outperformed other models in low-income (RMSE: 44.61, MAE: 35.24, MARE: 0.04, and R2: 0.94), medium-income (RMSE: 35.73, MAE: 28.42, MARE: 0.05, and R2: 0.92), and high-income (RMSE: 3.28, MAE: 2.69, MARE: 0.11, and R2: 0.84) categories.

To further illustrate the capabilities of the models, Fig. 3 (a–e) provides visual comparisons of their performance in the studied parameters. The figures offer a comprehensive overview of how each model fares in terms of forecast accuracy and efficiency across the different parameters examined. These results underscore the effectiveness of the RF model in modeling and predicting the occurrence of occupational accidents at various income levels. The superior performance of the M5 Tree model in predicting price index and the overall strong performance of the RF model validate their potential for informing decision-making processes related to accident prevention and workplace safety. The findings from this study contribute to the existing body of knowledge by providing insights into the comparative performance of different machine learning models and their suitability for predicting occupational accidents in relation to economic indicators.

#### 4.2.2. Predictions based on selected models

The predictions for the PI (using the M5 model), II (using the RF model), and BI (using the RF model) are depicted in Fig. 4. These predictions provide valuable insights into the expected values of these economic indicators for March 2021.

According to the findings, the predicted inflation index for March 2021 is 12.05, indicating a projected increase in the overall price levels compared to the base period. The price index is forecasted to reach 173.78, representing a potential rise in the average prices of goods and services. Additionally, the estimated monthly base income is projected to be approximately 2,221,000 Tomans (equivalent to about $96).

These predictions offer valuable information for policymakers, businesses, and individuals to anticipate and plan for the economic conditions in the near future. By utilizing machine learning models, such as M5 for price index prediction and RF for inflation index and basic income prediction, these forecasts can support decision-making processes, budgeting, and resource allocation strategies.

It is important to note that these predictions are based on the available data and the assumptions underlying the models used. As with any forecasting exercise, there is inherent uncertainty, and actual values

**Table 2**
Descriptive analysis of economic & accident data in the study population.

| | Parameter | Entire data (2007–2017) | Training set (2007–2015) | Test set (2016–2017) |
|---|---|---|---|---|
| Frequency of occupational accidents | | | | |
| Low income | N | 120 | 96 | 24 |
| | Sum | 109 868 | 91 605 | 18 263 |
| | Minimum | 536 | 556 | 536 |
| | Maximum | 1412 | 1412 | 941 |
| | Median | 900 | 955 | 761 |
| | Mean | 915.57 | 954.22 | 760.95 |
| | Standard deviation | 171.43 | 163.82 | 100.18 |
| Medium income | N | 120 | 96 | 24 |
| | Sum | 78 860 | 64 933 | 13 927 |
| | Minimum | 393 | 394 | 393 |
| | Maximum | 907 | 907 | 722 |
| | Median | 643 | 692 | 584.5 |
| | Mean | 657.17 | 676.38 | 580.29 |
| | Standard deviation | 118.86 | 118.72 | 84.67 |
| High income | N | 120 | 96 | 24 |
| | Sum | 3170 | 2532 | 638 |
| | Minimum | 11 | 12 | 11 |
| | Maximum | 41 | 41 | 39 |
| | Median | 27 | 26 | 28 |
| | Mean | 26.41 | 26.37 | 26.58 |
| | Standard deviation | 6.46 | 6.57 | 6.13 |
| Economic variables | | | | |
| Inflation index | N | 120 | 96 | 24 |
| | Sum | 2305.90 | 2021.6 | 284.30 |
| | Minimum | 8.60 | 8.80 | 8.60 |
| | Maximum | 40.40 | 40.40 | 15.60 |
| | Median | 17.5 | 20.45 | 11.55 |
| | Mean | 19.21 | 21.05 | 11.85 |
| | Standard deviation | 8.33 | 8.23 | 2.85 |
| Price index | N | 120 | 96 | 24 |
| | Sum | 6578.10 | 4266.50 | 2300 |
| | Minimum | 19.70 | 19.70 | 88.50 |
| | Maximum | 105.90 | 87.40 | 105.90 |
| | Median | 45 | 37.20 | 94.85 |
| | Mean | 54.81 | 44.55 | 95.85 |
| | Standard deviation | 27.89 | 20.90 | 4.93 |
| Basic income (Toman) | Minimum | 183 000 | 183 000 | 712 425 |
| | Maximum | 812 166 | 608 910 | 812 166 |
| | Median | 360 150 | 296 910 | 762 295 |
| | Mean | 427 064.1 | 343 256.25 | 762 295 |
| | Standard deviation | 219 407.6 | 145 077.93 | 70 527.53 |

**Fig. 2.** Occupational accident occurrences based on the income levels.